

Research Memo Template

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Analytical Or Research Question

Does a person's number of days they used public transportation per month on average over the last 3 months predict their composite communication apprehension score from responses to questions about their confidence in conversations?

Answer, Response, & Summary of Results

Yes. The number of days that a person uses public transportation in a week is a strong predictor of their composite communication apprehension score from questions related to their confidence in conversation.

```
options(conflicts.policy = "depends.ok")

library(tidyverse)
library(psych, exclude = c("%+%", "alpha"))
library(skimr)
library(Matrix, exclude = c("expand", "pack", "unpack"))
library(lme4)
library(kableExtra, exclude = "group_rows")
library(car, exclude = c("recode", "some", "logit"))
library(ggcorrplot)
library(broom)
library(skimr)
library(effects)

theme_set(theme_classic())

path_data <- "data"

data_c <- read_csv("C:/Users/hardy/OneDrive/Documents/capstone/data/PRCAQualtricsExport_FileC.csv", show_col_types = FALSE)

data_c <- data_c |>
  mutate(Q2 = recode(Q2, "Strongly disagree" = 5, "Somewhat disagree" = 4, "Neither agree nor disagree" = 3, "I agree" = 2, "Strongly agree" = 1),
         Q4 = recode(Q4, "Strongly disagree" = 5, "Somewhat disagree" = 4, "Neither agree nor disagree" = 3, "I agree" = 2, "Strongly agree" = 1),
         Q6 = recode(Q6, "Strongly disagree" = 5, "Somewhat disagree" = 4, "Neither agree nor disagree" = 3, "I agree" = 2, "Strongly agree" = 1),
         Q14 = recode(Q14, "Strongly disagree" = 5, "Somewhat disagree" = 3, "Neither agree nor disagree" = 2, "I agree" = 1, "Strongly agree" = 0),
         Q16 = recode(Q16, "Strongly disagree" = 5, "Somewhat disagree" = 3, "Neither agree nor disagree" = 2, "I agree" = 1, "Strongly agree" = 0),
         Q17 = recode(Q17, "Strongly disagree" = 5, "Somewhat disagree" = 4, "Neither agree nor disagree" = 3, "I agree" = 2, "Strongly agree" = 1),
         Q1 = recode(Q1, "Strongly disagree" = 1, "Somewhat disagree" = 2, "Neither agree nor disagree" = 3, "I agree" = 4, "Strongly agree" = 5),
         Q3 = recode(Q3, "Strongly disagree" = 1, "Somewhat disagree" = 2, "Neither agree nor disagree" = 3, "I agree" = 4, "Strongly agree" = 5),
         Q5 = recode(Q5, "Strongly disagree" = 1, "Somewhat disagree" = 2, "Neither agree nor disagree" = 3, "I agree" = 4, "Strongly agree" = 5),
         Q13 = recode(Q13, "Strongly disagree" = 1, "Somewhat disagree" = 2, "Neither agree nor disagree" = 3, "I agree" = 4, "Strongly agree" = 5),
         Q15 = recode(Q15, "Strongly disagree" = 1, "Somewhat disagree" = 1, "Neither agree nor disagree" = 2, "I agree" = 3, "Strongly agree" = 4),
         Q18 = recode(Q18, "Strongly disagree" = 1, "Somewhat disagree" = 1, "Neither agree nor disagree" = 2, "I agree" = 3, "Strongly agree" = 4))
```

```
data_c <- data_c |>
  mutate(Q_Composite = ((Q1 + Q2 + Q3 + Q4 + Q5 + Q6 + Q13 + Q14 + Q15 + Q16 + Q17 + Q18)))
print(min(data_c$Q_Composite, na.rm = TRUE))
```

```
[1] 12
```

```
print(max(data_c$Q_Composite, na.rm = TRUE))
```

```
[1] 60
```

```
data_c |>
  group_by(Q26) |>
  summarize(mean_comp = mean(Q_Composite, na.rm = TRUE), sd_comp = sd(Q_Composite, na.rm = TRUE), n = n())
  ungroup()
```

```
# A tibble: 6 x 4
```

Q26	mean_comp	sd_comp	n
<chr>	<dbl>	<dbl>	<int>
1 0-1 days a month	29.0	10.4	40
2 2-4 days a month	27.6	10.7	57
3 4-8 days a month	26.0	9.56	52
4 8 or more days a month	27.5	11.4	67
5 Never	32.2	13.0	55
6 <NA>	NaN	NA	2

```
data_c <- data_c |>
  mutate(Q26_num = case_match(Q26, "Never" ~ 0, "0-1 days a month" ~ 1, "2-4 days a month" ~ 2, "2-4 days a month" ~ 3, "8 or more days a month" ~ 4, "NA" ~ NA))
m_1 <- lm(Q_Composite ~ Q26_num, data = data_c)
summary(m_1)
```

Call:

```
lm(formula = Q_Composite ~ Q26_num, data = data_c)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.001	-8.650	-3.099	7.451	30.803

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	31.0014	1.2226	25.357	<2e-16 ***
Q26_num	-1.2010	0.4726	-2.541	0.0116 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.18 on 258 degrees of freedom

(13 observations deleted due to missingness)
 Multiple R-squared: 0.02442, Adjusted R-squared: 0.02064
 F-statistic: 6.458 on 1 and 258 DF, p-value: 0.01163

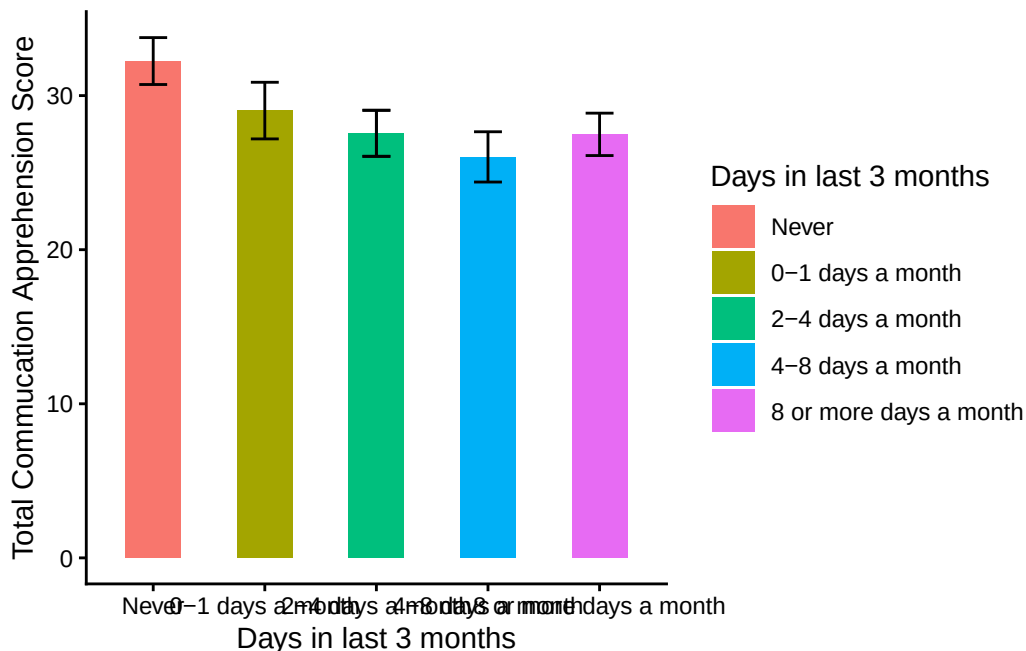
```
data_c <- data_c |>
  mutate(days_str = as.factor(Q26))

m_graph <- lm(Q_Composite ~ days_str, data = data_c)

x <- expand.grid(days_str = c("Never", "0-1 days a month", "2-4 days a month", "2-4 days a month", "4-8 days a month", "8 or more days a month"))

preds <- predict(m_graph, x, se.fit = TRUE) |>
  as_tibble() |>
  mutate(upper = fit + se.fit,
         lower = fit - se.fit) |>
  bind_cols(x)

ggplot(preds, aes(x = days_str, y = fit, fill = days_str)) +
  geom_bar(stat = "identity",
          position = position_dodge(width = .5),
          width = .5) +
  geom_errorbar(aes(ymin = lower, ymax = upper),
               position = position_dodge(width = .5),
               width = .25) +
  labs(x = "Days in last 3 months",
       y = "Total Communication Apprehension Score",
       fill = "Days in last 3 months")
```



This graph shows the gradual progression in total communication apprehension across responses to the number of days a person has used public transportation per month over the last 3 months. In general, the trend as the number of days public transportation was used per month over the last 3 months is a decrease in total communication apprehension score, with the exception of an excess of 8 or more days per month over the last 3 months.

```
pooled_sd <- sqrt(((40-1)*10.385584^2 + (67-1)*11.438454^2) / (40+67-2))  
print(pooled_sd)
```

```
[1] 11.0591
```

```
cohen_d = (29.02703 - 27.48485)/pooled_sd  
print(cohen_d)
```

```
[1] 0.139449
```

The mean composite communication apprehension scores of the group that never uses public transportation and the group that uses public transportation more than 8 days per month over the last three months differ by 0.14 standard deviations of both groups pooled. This implies that there is a moderately significant difference between their average composite communication apprehension scores.

Remaining Questions + Uncertainties

It seems clear that the number of days a participant has used public transportation per month over the last 3 months reliably predicts their composite communication apprehension score. However, it is not yet clear how other variables can influence these predictions. How can variables like age, gender, and employment status affect these results?

References